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QUESTION BANK PHYSICS 2K23 (SEM-I)

LASER

Q) Full form of laser
Q) Define the term :
Absorbtion , Emission , Spontaneous , Stimulated Emission
Q) Diff Bet :
LASER and Ordinary Light
Spontaneous and Stimulated Emission
Ruby laser vs He-Ne laser
Q) Short notes on
Pumping and its Techniques and Scheme .
Optical Resonator
Application of laser
Q) Working of Ruby and He-Ne laser
Q) Diagram :
RUBY , HE- NE , Metastable state ,
Energy level diagrams of lasers

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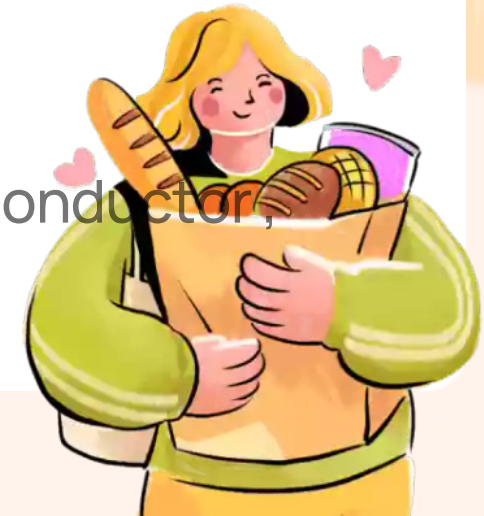
OPTICAL FIBRE

Q) Define the term :
Optical fiber, Critical angle , Acceptance angle
Q) Optical fiber work on which principle ?
Q) Explain Structure of optical fibre , Optical Resonator.
Q) Short notes on :
Types of losses , Type of optical fibre
Q) Relation bet N.A and Fraction Refractive index
Q) Expression for an Accptance angle For the signal propagating throught an optical fibre
Q) Applications any two

SEMICONDUCTOR

Q) Define the term :
Semiconductors and its type , Conductor ,



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Conductivity, Fermi Energy

Q) Diff bet :

Conductor, semiconductor and Insulator , P type vs N type , Intrinsic vs Extrinsic.

Q) Explain formula for Fermi Energy in Intrinsic Semiconductors ?

Q) what is p-n junction diode . Explain the formation of Depletion regions.

Q) what Is hall effect , drive an expression for hall coefficient

Q) Explain various type of logic gates in short

Q) give four Applications of hall effect

CRYSTAL STRUCTURES AND NANOSCIENCE

Q) Define the term :

Space lattice , basis , unit cell , bravis lattice.

Q) Explain Cordination number in Scc , Bcc , Fcc.

Q) Write 7 crystals system with there primitives , Interfacial angles and Bravais Lattice points.

Q) Write The fourteen Bravais lattices

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Q) Define atomic radius and packing fraction. Calculate the atomic radii and packing fractions for body centered and face centered Cubic Unit Cell.

Q) Find Miller indices of given planes (110) (211) (012) (100).

Q) Draw a plane having millar indices (100) (121) (010) (001) (111).

Q) Derive Bragg,s law for X-ray diffraction in crystals. State any one application of it.